

G-quadruplex-based aptamers against protein targets in therapy and diagnostics.

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Nucleic acid aptamers are single-stranded DNA or RNA molecules identified to recognize with high affinity specific targets including proteins, small molecules, ions, whole cells and even entire organisms, such as viruses or bacteria. They can be identified from combinatorial libraries of DNA or RNA oligonucleotides by SELEX technology, an in vitro iterative selection procedure consisting of binding (capture), partitioning and amplification steps. Remarkably, many of the aptamers selected against biologically relevant protein targets are G-rich sequences that can fold into stable G-quadruplex (G4) structures. Aiming at disseminating novel inspiring ideas within the scientific community in the field of G4-structures, the emphasis of this review is placed on: 1) recent advancements in SELEX technology for the efficient and rapid identification of new candidate aptamers (introduction of microfluidic systems and next generation sequencing); 2) recurrence of G4 structures in aptamers selected by SELEX against biologically relevant protein targets; 3) discovery of several G4-forming motifs in important regulatory regions of the human or viral genome bound by endogenous proteins, which per se can result into potential aptamers; 4) an updated overview of G4-based aptamers with therapeutic potential and 5) a discussion on the most attractive G4-based aptamers for diagnostic applications. This article is part of a Special Issue entitled "G-quadruplex" Guest Editor: Dr. Concetta Giancola and Dr. Daniela Montesarchio.

Constructive Prediction of Potential RNA Aptamers for a Protein Target.

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Aptamers are short single-stranded nucleic acids that bind to target molecules with high affinity and selectivity. Aptamers are generally identified in vitro by performing SELEX (systematic evolution of ligands by exponential enrichment). Complementing the SELEX process, several computational methods have been proposed in the search for aptamers. However, many of these methods cannot be applied for finding new aptamers, either because they are classifiers for determining whether an RNA and protein interact with each other, or because they are limited to a specific target only. Hence, we developed a new random forest (RF) model for finding potential RNA aptamers for a protein target. From an extensive analysis of protein-RNA complexes including RNA aptamers-protein complexes, we identified key features of interacting RNA and protein molecules, and structural constraints on RNA aptamers. The potential RNA aptamers predicted by our method reveal similar secondary and protein-binding structures as the actual RNA aptamers. The RF model showed a reliable performance in both cross validations and independent testing. The key features of interacting RNA and protein molecules and the structural constraints identified in our study were effective in finding potential aptamers for a protein target. Although preliminary, our results are promising, and we believe this approach will be useful in reducing time and money spent on in vitro experiments by substantially limiting the size of the initial pool of nucleic acid sequences.

A binary Cy3 aptamer probe composed of folded modules.

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Aptamers are short single-stranded DNA or RNA sequences that are selected in vitro based on their high

affinity to a target molecule. Dye-binding aptamers are promising tools for real-time detection of not only DNA or RNA sequences but also proteins of interest both in vitro and in vivo. In this study, we aimed to isolate an RNA aptamer to Cy3, a widely used, membrane-permeant, and nontoxic fluorescent cyanine dye. Extensive selection of affinity RNA molecules to Cy3 yielded a unique sequence aptamer named Cy3_apt. The selected Cy3_apt was 83 nucleotides long and successfully shortened to 49 nucleotides long with increased affinity to Cy3 by multiple base changes. The shortest Cy3_apt is composed of two separate hairpin modules that are required for the affinity to Cy3 as monitored by the surface plasmon resonance (SPR) assay. Also, the fluorescence of Cy3 increased on binding to Cy3_apt. The two modules of Cy3_apt, when detached from each other, functioned as a binary aptamer probe. We demonstrate that the binary Cy3_apt probe is applicable to the detection of target oligonucleotides or RNA-RNA interaction by tagging with target sequences. This binary probe consists of two folded modules, referred to as a folded binary probe.

Ligand Guided Selection (LIGS) of Artificial Nucleic Acid Ligands against Cell Surface Targets.

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With the success of RNA-based therapeutic drugs, the demand has increased for sophisticated nucleic-acid-based targeting agents. Nucleic acid aptamers (NAAs), in this regard, represent a suitable class of molecules with synthetic versatility. Aptamers are composed of single-stranded RNA/DNA/XNA molecules, which can be identified using a method called systematic evolution of ligands by exponential enrichment (SELEX) against any molecule. This Spotlight summarizes the recent introduction of ligand guided selection (LIGS), which will permit the identification of a wide range of functional aptamers against complex targets such as cell surface receptors while maintaining their native functional state. Aptamers identified from LIGS will allow researchers to develop aptamers in biomedicine as low-cost, stable therapeutic agents and diagnostic molecules or biochemical devices.

Nucleic Acid Aptamer-Based Biosensors: A Review.

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Aptamers, short strands of either DNA, RNA, or peptides, known for their exceptional specificity and high binding affinity to target molecules, are providing significant advancements in the field of health. When seamlessly integrated into biosensor platforms, aptamers give rise to aptasensors, unlocking a new dimension in point-of-care diagnostics with rapid response times and remarkable versatility. As such, this review aims to present an overview of the distinct advantages conferred by aptamers over traditional antibodies as the molecular recognition element in biosensors. Additionally, it delves into the realm of specific aptamers made for the detection of biomarkers associated with infectious diseases, cancer, cardiovascular diseases, and metabolomic and neurological disorders. The review further elucidates the varying binding assays and transducer techniques that support the development of aptasensors. Ultimately, this review discusses the current state of point-of-care diagnostics facilitated by aptasensors and underscores the immense potential of these technologies in advancing the landscape of healthcare delivery.

Recent focus in non-SELEX-computational approach for de novo aptamer design: A mini review.

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Aptamers, single-stranded nucleic acids that bind to specific targets with high affinity and specificity, hold significant promise in various biomedical and biotechnological applications. The traditional method of aptamer selection, SELEX (Systematic Evolution of Ligands by EXponential Enrichment) takes a lot of work and time. Recent advancements in computational methods have revolutionized aptamer design, offering efficient and effective alternatives. This review examines recent advances in non-SELEX and de novo aptamer design methods, such as Making Aptamers without SELEX (MAWS), AptaLoop, AptaDiff, RNAGEN, RaptGen, Apta-MCTS, UltraSelex, and Torkamanian-Afshar model. These computer methods utilize bioinformatics, machine learning, and molecular modeling to generate high-affinity aptamers, eliminating the need for multiple selection steps in vitro or in vivo. We provide a comprehensive analysis of each method's performance, including binding affinity, specificity, and stability, and discuss their practical applications in diagnostics, therapeutics, and environmental monitoring. Furthermore, we highlight the strengths and limitations of computational methods against the traditional one. The potential challenges, future directions, and emerging technologies were also presented. This review underscores the transformative impact of computational aptamer design on research and industry, paving the way for rapid and cost-effective development of aptamer-based technologies.

Systematic bio-fabrication of aptamers and their applications in engineering biology.

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Aptamers are single-stranded DNA or RNA molecules that have high affinity and selectivity to bind to specific targets. Compared to antibodies, aptamers are easy to in vitro synthesize with low cost, and exhibit excellent thermal stability and programmability. With these features, aptamers have been widely used in biology and medicine-related fields. In the meantime, a variety of systematic evolution of ligands by exponential enrichment (SELEX) technologies have been developed to screen aptamers for various targets. According to the characteristics of targets, customizing appropriate SELEX technology and post-SELEX optimization helps to obtain ideal aptamers with high affinity and specificity. In this review, we first summarize the latest research on the systematic bio-fabrication of aptamers, including various SELEX technologies, post-SELEX optimization, and aptamer modification technology. These procedures not only help to gain the aptamer sequences but also provide insights into the relationship between structure and function of the aptamers. The latter provides a new perspective for the systems bio-fabrication of aptamers. Furthermore, on this basis, we review the applications of aptamers, particularly in the fields of engineering biology, including industrial biotechnology, medical and health engineering, and environmental and food safety monitoring. And the encountered challenges and prospects are discussed, providing an outlook for the future development of aptamers.

Application of Electrochemical Aptasensors toward Clinical Diagnostics, Food, and Environmental Monitoring: Review.

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Aptamers are synthetic bio-receptors of deoxyribonucleic acid (DNA) or ribonucleic acid (RNA) origin selected by the systematic evolution of ligands (SELEX) process that bind a broad range of target analytes with high affinity and specificity. So far, electrochemical biosensors have come up as a simple and sensitive method to utilize aptamers as a bio-recognition element. Numerous aptamer based sensors have been developed for clinical diagnostics, food, and environmental monitoring and several other applications are under development. Aptasensors are capable of extending the limits of current analytical techniques in clinical diagnostics, food, and environmental sample analysis. However, the potential applications of aptamer based electrochemical biosensors are unlimited; current applications are observed in the areas of food toxins, clinical biomarkers, and pesticide detection. This review attempts to enumerate the most representative examples of research progress in aptamer based electrochemical biosensing principles that have been developed in recent years. Additionally, this account will discuss various current developments on aptamer-based sensors toward heavy metal detection, for various cardiac biomarkers, antibiotics detection, and also on how the aptamers can be deployed to couple with antibody-based assays as a hybrid sensing platform. Aptamers can be used in various applications, however, this account will focus on the recent advancements made toward food, environmental, and clinical diagnostic application. This review paper compares various electrochemical aptamer based sensor detection strategies that have been applied so far and used as a state of the art. As illustrated in the literature, aptamers have been utilized extensively for environmental, cancer biomarker, biomedical application, and antibiotic detection and thus have been extensively discussed in this article.

Single-stranded DNA (ssDNA) production in DNA aptamer generation.

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The discovery that synthetic short chain nucleic acids are capable of selective binding to biological targets has made them to be widely used as molecular recognition elements. These nucleic acids, called aptamers, are comprised of two types, DNA and RNA aptamers, where the DNA aptamer is preferred over the latter due to its stability, making it widely used in a number of applications. However, the success of the DNA selection process through Systematic Evolution of Ligands by Exponential Enrichment (SELEX) experiments is very much dependent on its most critical step, which is the conversion of the dsDNA to ssDNA. There is a plethora of methods available in generating ssDNA from the corresponding dsDNA. These include asymmetric PCR, biotin-streptavidin separation, lambda exonuclease digestion and size separation on denaturing-urea PAGE. Herein, different methods of ssDNA generation following the PCR amplification step in SELEX are reviewed.

Evolution of Complex Target SELEX to Identify Aptamers against Mammalian Cell-Surface Antigens.

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The demand has increased for sophisticated molecular tools with improved detection limits. Such molecules should be simple in structure, yet stable enough for clinical applications. Nucleic acid aptamers (NAAs) represent a class of molecules able to meet this demand. In particular, aptamers, a class of small nucleic acid ligands that are composed of single-stranded modified/unmodified RNA/DNA molecules, can be evolved from a complex library using Systematic Evolution of Ligands by EXponential

enrichment (SELEX) against almost any molecule. Since its introduction in 1990, in stages, SELEX technology has itself undergone several modifications, improving selection and broadening the repertoire of targets. This review summarizes these milestones that have pushed the field forward, allowing researchers to generate aptamers that can potentially be applied as therapeutic and diagnostic agents.
